

Qatar-4b

R-2710

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_250619 · created 2026-09-01T01:05:30+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460845.9077 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.028
Transit depth (Rp/Rs)^2	3.21 +/- 1.01 [%]
Orbital Inclination (inc)	89.56 +/- 1.58
Airmass coefficient 1 (a1)	1.056 +/- 0.039
Airmass coefficient 2 (a2)	-0.031 +/- 0.016
Scatter in the residuals of the lightcurve fit is	3.68 %
Best Comparison Star	#2 - [309, 323]
Optimal Aperture	3.72
Optimal Annulus	10.94
Transit Duration (day)	0.0944 +/- 0.0032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.179 $\pm$ 0.028 vs 0.138 (deviation 1.46 σ)
Duration fit vs published	0.0944 d vs 0.0892 d (deviation 1.64 σ)
Mid-time O-C	1.0 min
Depth SNR	6.4
Residual scatter	3.68 %

Light curve

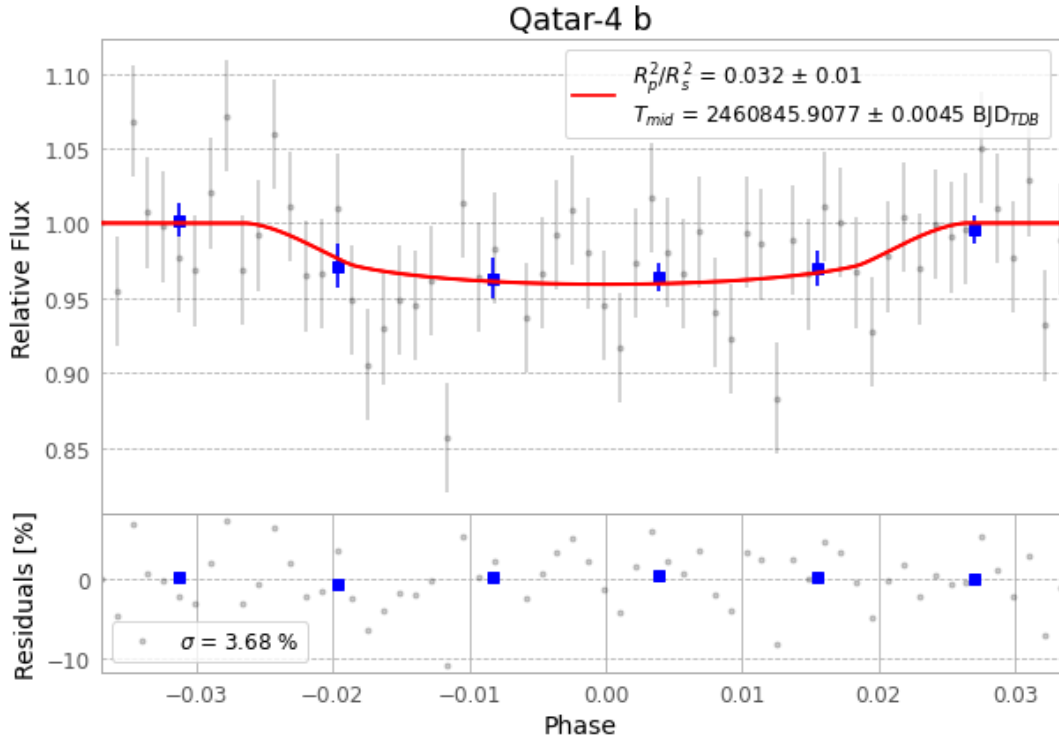


Figure 1 — FinalLightCurve_Qatar-4 b_2025-06-19.png (SHA-256 26a2e6c014739f59...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [345, 154], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 1a3adb75a1d15f9b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

62 FITS frames (41 MB), Qatar-4250619081216.FITS → Qatar-4250619111515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-250619.log (44b9544c3bfd...), FinalLightCurve_Qatar-4 b_2025-06-19.png (26a2e6c01473...), FinalLightCurve_Qatar-4 b_2025-06-19.csv (5791f789bd52...)

Compute resources

Wall time 113.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 01:05Z.